

Week 6 Deliverable:
Detailed Rocker-Bogie Suspension CAD
Assembly
and Geometry Refinement

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Conceptual Design of a Lunar Regolith Transport Rover

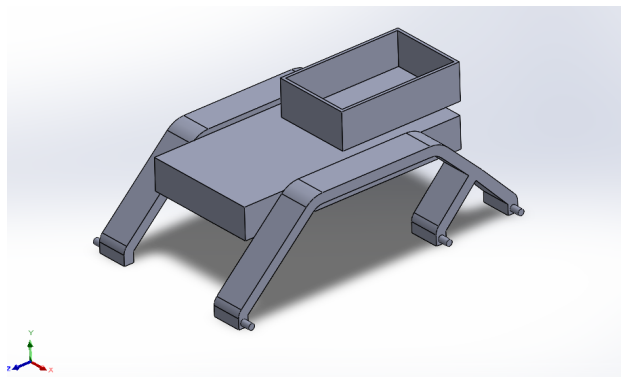


Figure 1: Week 4 preliminary block model used as the starting point for the detailed Week 6 suspension design.

1 Week 6 Objective

The objective of Week 6 was to replace the simplified rocker-bogie geometry from the Week 4 block model with a detailed and mechanically organized CAD assembly. The Week 4 model represented the suspension using basic links, pivots, and wheel cylinders so that the initial rover proportions could be established [2]. Week 6 develops these simplified components into individual rocker and bogie members, reinforced joints, rectangular beam sockets, wheel mounts, pivot housings, fastener holes, and a complete six-wheel mobility assembly.

The Week 6 deliverable is intentionally limited to the rocker-bogie mobility system. The final chassis, dump bed, battery enclosure, electronics packaging, sensors, and complete rover body are not included in this deliverable. Those systems will be integrated during the later full-rover assembly phase.

The primary Week 6 goals were:

1. Convert an imported rover reference assembly into editable CAD components.
2. Replace the original circular suspension rods with rectangular hollow structural members.
3. Design reinforced rectangular sockets for the suspension members.
4. Create detailed rocker-to-bogie pivot joints.
5. Adapt the previously designed lunar wheels to detailed wheel mounts.
6. Refine the suspension member lengths and assembly proportions.
7. Introduce a modest rear-wheel lateral offset for stability.
8. Assemble the complete left and right rocker-bogie mobility systems.
9. Document the geometry changes from the Week 4 block-model baseline.

2 Prior Rover Design Baseline

The lunar rover concept was established in Week 3 as a semi-autonomous transport vehicle designed to carry loose lunar regolith from a collection area to a habitat shielding area [1]. The rover uses six wheels, a rocker-bogie suspension, a rear dump bed, and independent wheel-drive assemblies.

The Week 4 and Week 5 parameters carried into the detailed Week 6 suspension design are summarized in Table 1.

Table 1: Rover baseline used for the Week 6 mobility assembly [2, 3].

Parameter	Baseline value
Regolith payload mass	340 kg
Estimated dry rover mass	650 kg
Loaded rover mass	990 kg
Lunar gravitational acceleration	1.62 m/s ²
Loaded lunar rover weight	1603.8 N
Approximate normal load per wheel	267.3 N
Number of wheels	6
Nominal wheel diameter	0.50 m
Nominal wheel radius	0.25 m
Original adjacent-wheel spacing target	0.9 m to 1.0 m
Original total wheelbase target	2.0 m to 2.3 m
Original track-width target	Approximately 1.8 m
Ground-clearance target	0.30 m to 0.40 m
Continuous output torque per wheel	40 N m to 50 N m
Peak output torque per wheel	80 N m to 100 N m
Wheel output speed	Approximately 8 rpm

The Week 3 concept identified several structural concerns that directly influenced the Week 6 redesign. The suspension arms should not be excessively long or thin, and reinforced rectangular or box-section members were recommended instead of thin circular rods. Reinforced pivot brackets, gussets, double-shear mounting, protected joints, and dust-resistant wheel mounts were also identified as important design requirements [1].

3 Rocker-Bogie Suspension Architecture

The rocker-bogie suspension is divided into a differential connection, one rocker assembly on each side of the rover, and one bogie assembly on each side. The rocker connects the front wheel to the rover differential and to the bogie, while the bogie connects the middle and rear wheels to the rocker [7].

The configuration allows the wheel positions to change relative to one another as the rover crosses uneven terrain. The rocker-bogie architecture has extensive planetary-rover heritage

and was selected for NASA Mars rovers because of its terrain-following ability, stability, and capability to limit rover-body tilt when individual wheels encounter obstacles [5, 8].

The intended structural load path for the complete rover remains:

Regolith payload → Dump bed → Chassis → Main rocker pivots → Rocker and bogie members → Wheel (1)

The dump bed will eventually mount to the chassis rather than directly to the moving suspension members. This keeps the rocker and bogie links free to rotate while the chassis carries the primary payload and equipment loads.

4 CAD Reference Assembly and Attribution

A publicly available rover assembly associated with Abdul Azeez Mohammad's GrabCAD profile was downloaded and used as a visual and mechanical reference [4]. The imported assembly provided useful examples of:

- rocker and bogie organization,
- pivot-housing geometry,
- wheel-mount packaging,
- steering-mount geometry,
- differential-link placement,
- suspension articulation, and
- overall assembly organization.

The GrabCAD assembly was not used directly as the final rover design. The original model contained circular rods, dimensions, joints, wheel mounts, and wheels that did not match the current 340 kg payload rover. Individual parts were therefore scaled, split, moved, rebuilt, or replaced with new geometry.

The STEP assembly was initially opened through SOLIDWORKS 3D Interconnect. The top-level STEP connection was broken so the individual components could be saved as editable files. The imported parts then served as reference geometry during the Week 6 redesign.

5 Week 6 CAD Redesign Process

The detailed suspension was created through the following sequence:

1. The STEP assembly was converted into editable SOLIDWORKS components.

2. The original reference wheels were removed because a separate lunar wheel had already been designed during the mobility-development phase.
3. Circular suspension rods were replaced with rectangular hollow members.
4. Selected imported joint bodies were scaled to match the larger suspension members.
5. Circular arm connections were replaced with rectangular sockets.
6. Wheel mounts were resized and lengthened to provide adequate wheel and grouser clearance.
7. Through-bolt holes were added to secure the suspension members inside the sockets.
8. The central rocker-to-bogie connection was modeled as a rotating pivot rather than a rigid beam connection.
9. The left and right suspension assemblies were positioned around the rover center plane.
10. The suspension member lengths and rear-wheel lateral positions were refined after assembling the mobility system.

This method preserved useful mechanical concepts from the downloaded reference while substantially changing the dimensions, structural members, wheel interfaces, and connection geometry.

6 Rectangular Suspension Members

The redesigned suspension uses rectangular hollow members instead of the original circular rods. A common conceptual cross-section was selected for the rocker and bogie members:

$$h_{\text{beam}} \times b_{\text{beam}} = 100 \text{ mm} \times 50 \text{ mm}. \quad (2)$$

The preliminary beam wall thickness is:

$$t_{\text{beam}} = 5 \text{ mm}. \quad (3)$$

The resulting internal dimensions are approximately:

$$h_{\text{inner}} = 100 - 2(5) = 90 \text{ mm}, \quad (4)$$

$$b_{\text{inner}} = 50 - 2(5) = 40 \text{ mm}. \quad (5)$$

The 100 mm dimension is oriented within the main suspension bending plane. This gives the member greater bending stiffness than rotating the section so that the 50 mm dimension carries the main vertical bending load.

The rectangular member dimensions are preliminary conceptual dimensions. They have not yet been verified through detailed stress, buckling, fatigue, or finite-element analysis.

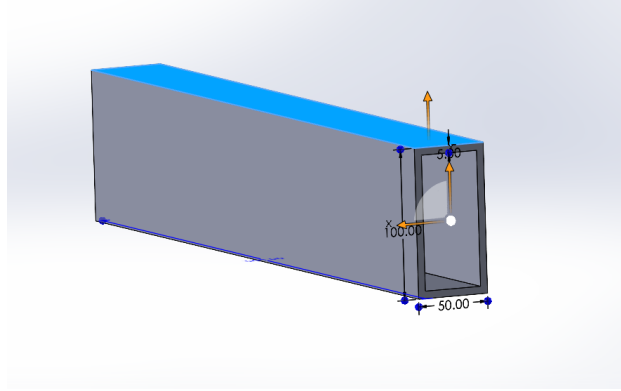


Figure 2: Rectangular hollow section used for the rocker and bogie members.

7 Final Week 6 Arm-Length Selection

The detailed joints, sockets, and wheel mounts add effective length beyond the straight beam sections. The beam extrusion lengths therefore cannot be treated as equal to the final wheel-center spacing.

Several longer arm configurations were initially tested during assembly. Those configurations produced excessive wheel spacing relative to the Week 4 block-model targets. The members were shortened through multiple CAD iterations until the suspension had more compact and visually consistent proportions.

The final Week 6 straight-member lengths are:

$$L_{\text{bogie,short}} = 450 \text{ mm}, \quad (6)$$

$$L_{\text{bogie,long}} = 500 \text{ mm}, \quad (7)$$

$$L_{\text{rocker}} = 800 \text{ mm}. \quad (8)$$

The short and long bogie members connect the middle and rear wheel-mount assemblies to the central bogie joint. The longer 800 mm member forms the primary rocker segment leading toward the front-wheel assembly.

The sum of the straight member lengths is:

$$L_{\text{members,total}} = 450 + 500 + 800 = 1750 \text{ mm}. \quad (9)$$

This value is not the rover wheelbase. The pivot housings, beam insertion depths, rectangular sockets, wheel mounts, and joint offsets all contribute additional distance between the wheel axes.

The final wheel-center spacing should therefore be measured directly from the completed neutral-position CAD assembly rather than estimated by adding the beam lengths.

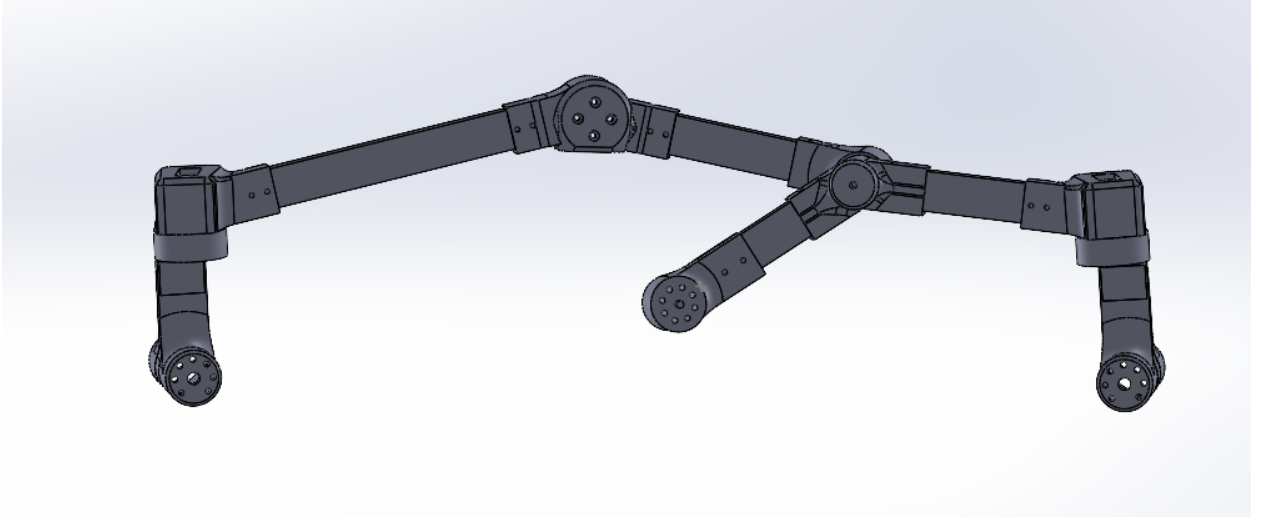


Figure 3: Dimensioned rocker-bogie member geometry used in the final Week 6 assembly.

Table 2: Final Week 6 suspension-member lengths.

Suspension member	Final length
Short bogie member	450 mm
Long bogie member	500 mm
Main rocker member	800 mm

8 Rectangular Beam Socket Design

Each rocker and bogie member enters a reinforced rectangular socket attached to a wheel mount or central pivot housing. The selected socket dimensions are:

$$h_{\text{socket,outer}} = 120 \text{ mm}, \quad (10)$$

$$b_{\text{socket,outer}} = 70 \text{ mm}, \quad (11)$$

$$h_{\text{socket,inner}} = 104 \text{ mm}, \quad (12)$$

$$b_{\text{socket,inner}} = 54 \text{ mm}. \quad (13)$$

The vertical clearance between the beam and socket is:

$$c_h = \frac{104 - 100}{2} = 2 \text{ mm} \quad (14)$$

on each side.

The lateral clearance is:

$$c_b = \frac{54 - 50}{2} = 2 \text{ mm} \quad (15)$$

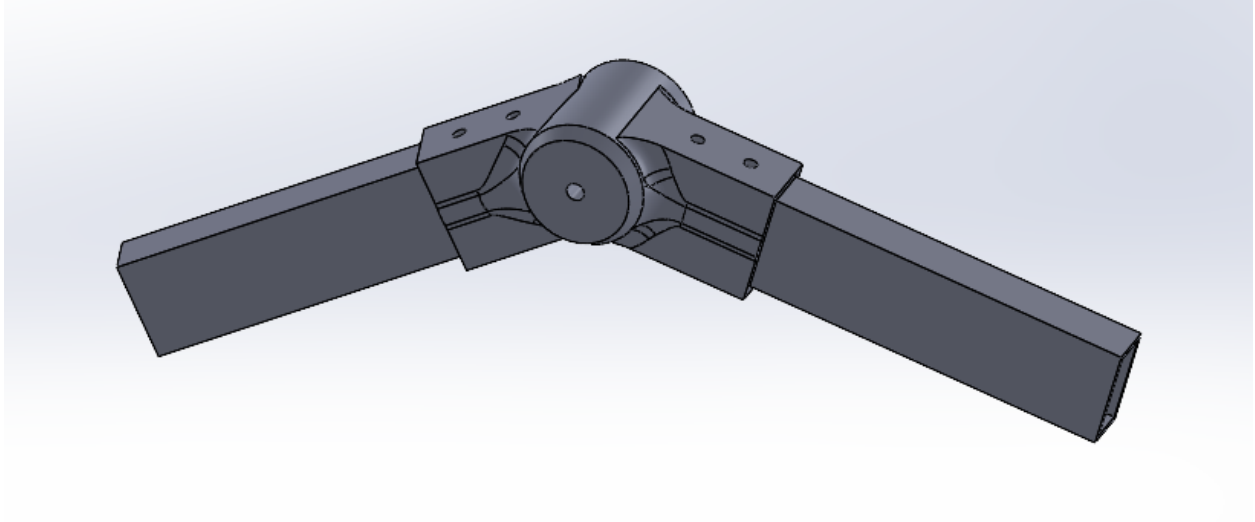


Figure 4: Section view of the rectangular beam socket and inserted suspension member.

on each side.

The nominal socket wall thickness is:

$$t_{\text{socket}} = \frac{120 - 104}{2} = \frac{70 - 54}{2} = 8 \text{ mm.} \quad (16)$$

The nominal beam engagement depth is approximately:

$$L_{\text{engagement}} \approx 125 \text{ mm to } 140 \text{ mm.} \quad (17)$$

The exact engagement depth varies slightly depending on the available material near the wheel mount or central pivot housing.

The rectangular opening stops before entering the wheel-bearing or central pivot geometry. Approximately 20 mm to 25 mm of solid material may be retained behind the beam as an internal stop when sufficient space is available.

9 Beam-to-Socket Fasteners

The suspension members are retained inside the sockets using through-bolts. The primary beam connections use:

- M12 bolts,
- approximately 13 mm clearance holes,
- washers,
- locking nuts,
- approximately 50 mm to 55 mm hole-center spacing, and

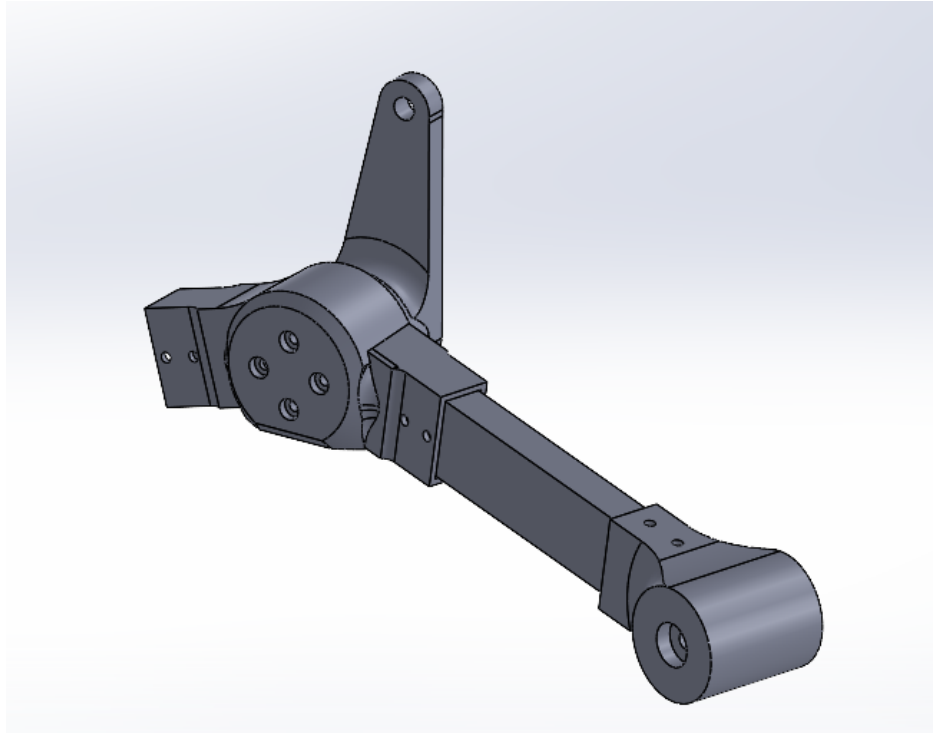


Figure 5: Beam-to-socket through-bolt arrangement.

- approximately 30 mm from the socket opening to the first hole center.

Each bolt passes through:

$$\text{near socket wall} \rightarrow \text{rectangular beam} \rightarrow \text{far socket wall.} \quad (18)$$

This creates a complete through-bolt connection rather than fastening only one socket wall.

Compact mounting regions with limited space may use M10 bolts with approximately 12 mm clearance holes. These smaller fasteners are used only where the available material or edge distance does not support the larger M12 connection.

The through-bolts prevent the beam from sliding or rotating inside the socket. The joint must still be checked later for bolt shear, bearing stress, hole tear-out, beam-wall crushing, and cyclic loading.

10 Rocker-to-Bogie Pivot Joint

The central rocker-to-bogie connection performs a different function from the rigid beam sockets. The beam-to-socket bolts prevent relative movement, while the central pivot must allow the bogie to rotate relative to the rocker.

The detailed pivot design includes:

- separate rocker-side and bogie-side components,

- a large circular central housing,
- rectangular beam sockets,
- an approximately 30 mm smooth pivot pin,
- a conceptual bushing bore of approximately 30.5 mm to 31 mm,
- an approximate bushing outside diameter of 40 mm to 45 mm,
- axial locating faces or spacers, and
- smaller cover-retaining fasteners.

A smooth pivot pin or shoulder-bolt arrangement is preferred over a fully threaded bolt. The smooth portion of the pin should pass through the rotating bearing or bushing surfaces. Any threaded retaining section should remain outside the main rotating contact region.

The smaller fasteners around the circular face are intended to represent cover or housing fasteners. They do not replace the primary central pivot pin and should not be treated as the main suspension load path.

The intended central joint load path is:

rocker member → rocker-side housing → pivot pin and bushing → bogie-side housing → bogie members.
(19)

11 Wheel-Mount Redesign

The wheel mounts were adapted from the imported reference geometry but were resized and reconstructed for the current lunar wheel and rectangular suspension members. NASA planetary rovers use dedicated wheel and steering assemblies that separate suspension structure from wheel-drive, bearing, and steering functions [7, 9].

The redesigned wheel mount contains:

- a rectangular socket for the 100 mm by 50 mm member,
- a rounded structural transition,
- a circular wheel-side mounting flange,
- a central drive or bearing interface,
- multiple wheel-flange bolt locations,
- an upper support structure,
- tire and grouser clearance, and
- beam-retaining through-bolt holes.

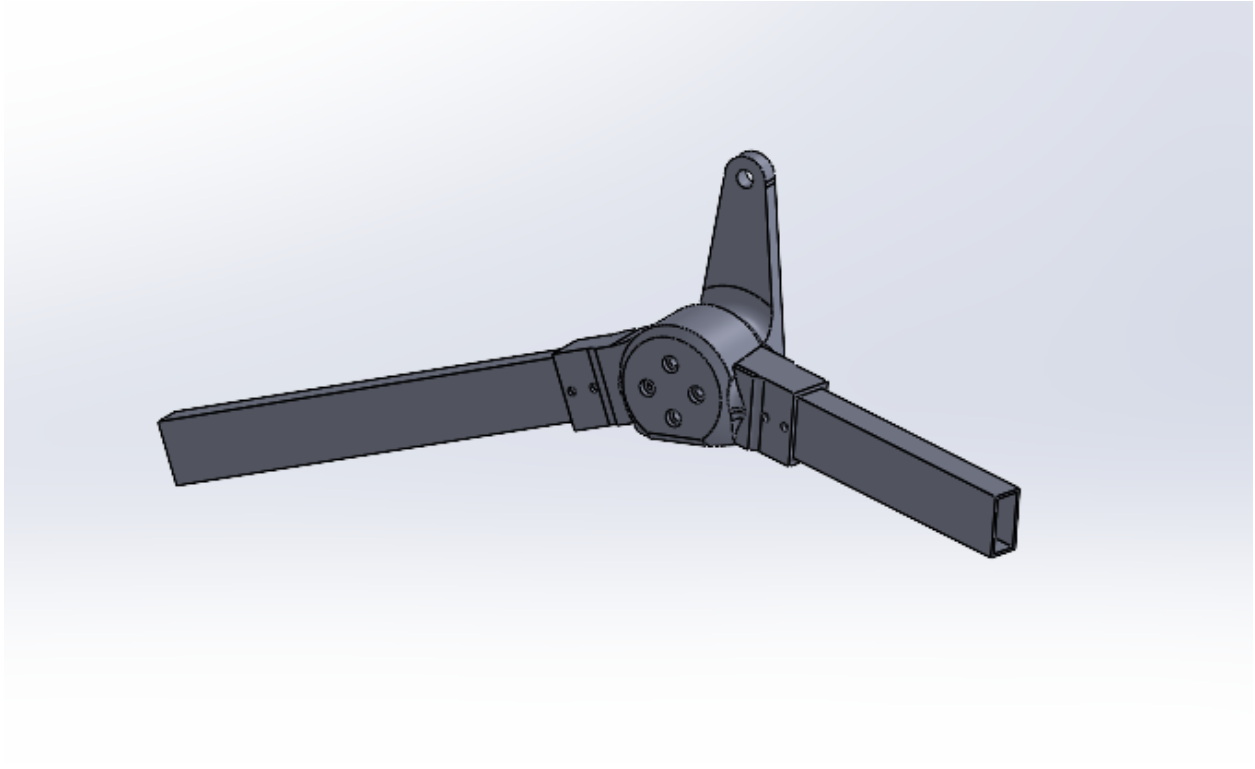


Figure 6: Detailed rocker-to-bogie pivot joint and rectangular member connections.

The upper section of the front and rear wheel-carrier geometry was lengthened using a split-body and move-body operation rather than nonuniformly scaling the entire mount. This preserved the circular holes, bearing interfaces, and wheel-flange geometry while increasing vertical clearance.

The conceptual minimum clearance between the tallest wheel grouser and the nearest stationary wheel-mount surface is:

$$c_{\text{wheel}} \approx 25 \text{ mm to } 40 \text{ mm.} \quad (20)$$

This clearance should be checked throughout the available wheel rotation, steering motion, and suspension articulation range.

12 Geometry Refinement From the Week 4 Model

The Week 4 block model used adjacent-wheel spacing targets of approximately 0.9 m to 1.0 m and a total wheelbase target of approximately 2.0 m to 2.3 m [2]. These values were preliminary and were established before detailed joints, sockets, wheel mounts, fastener edge distances, and beam-insertion lengths were modeled.

The first detailed suspension trials used longer straight members. After assembly, the wheel-center distances became larger than the original conceptual range. The members were shortened incrementally until the assembly became more compact.

The final Week 6 member lengths were set to:

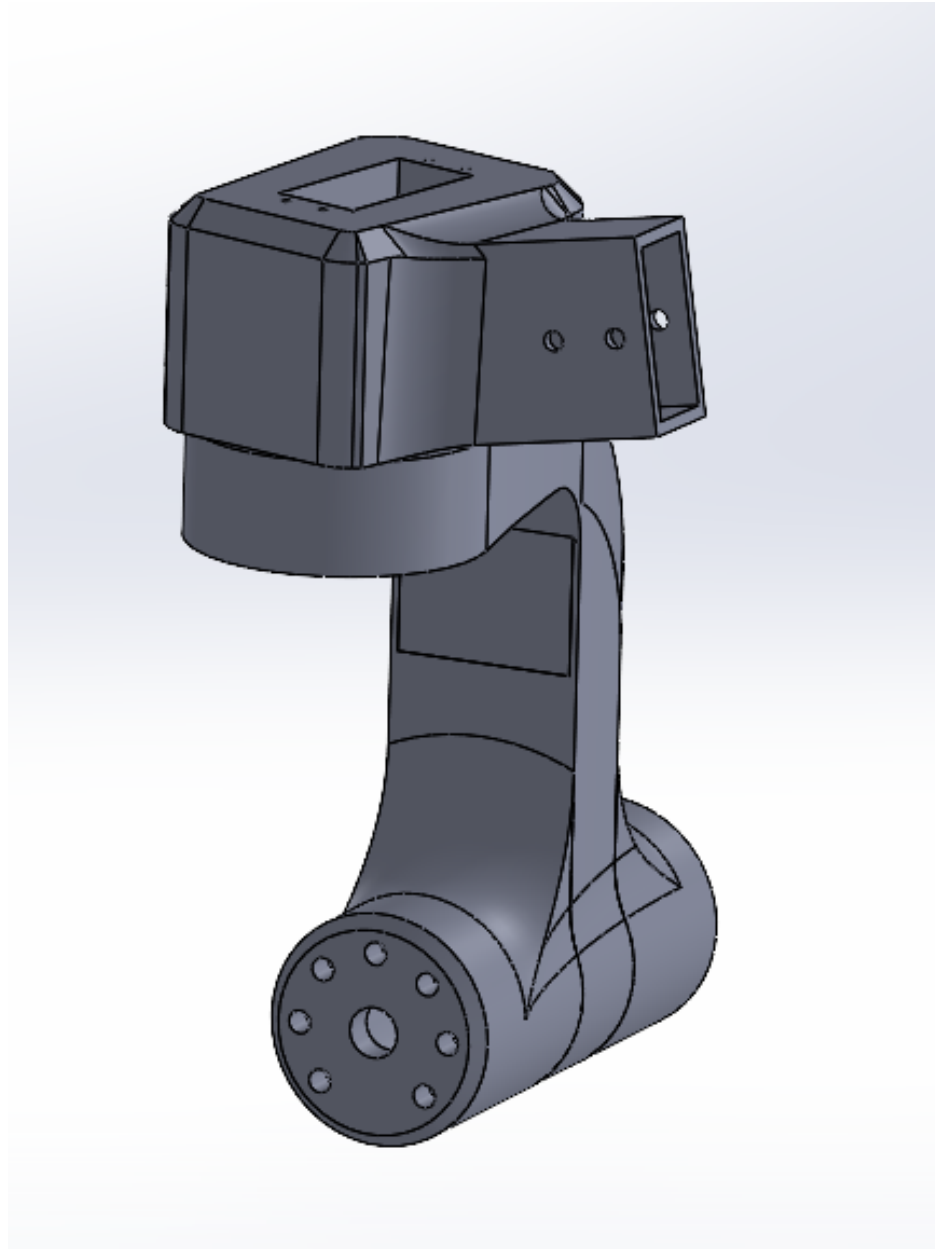


Figure 7: Redesigned wheel mount connected to a rectangular suspension member.

$$\boxed{L_{\text{bogie,short}} = 450 \text{ mm}, \quad L_{\text{bogie,long}} = 500 \text{ mm}, \quad L_{\text{rocker}} = 800 \text{ mm}} \quad (21)$$

These dimensions represent the straight rectangular members. They do not include the effective lengths of:

- the wheel mounts,
- rectangular sockets,
- beam insertion regions,
- central pivot housings,
- wheel-flange offsets, or
- joint clearances.

The final adjacent-wheel spacing must therefore be obtained from direct axis-to-axis measurement in the completed assembly.

Table 3 summarizes the main changes from the Week 4 block model.

Table 3: Week 4 baseline compared with the detailed Week 6 CAD design.

Geometry feature	Week 4 baseline	Week 6 refinement
Suspension representation	Simplified bars and pivots	Separate beams, sockets, wheel mounts, and pivot housings
Member section	Not finalized	100 mm by 50 mm hollow section
Preliminary wall thickness	Not finalized	Approximately 5 mm
Short bogie member	Not individually finalized	450 mm
Long bogie member	Not individually finalized	500 mm
Main rocker member	Not individually finalized	800 mm
Socket exterior	Not defined	120 mm by 70 mm
Socket interior	Not defined	104 mm by 54 mm
Primary socket fasteners	Not defined	M12 bolts with approximately 13 mm holes
Rocker-to-bogie pivot	Simplified pin	Approximately 30 mm smooth pivot pin
Rear-wheel lateral position	Common initial track-width assumption	Intentional outward rear-wheel offset

13 Rear-Wheel Lateral Offset

During the final Week 6 assembly refinement, the rear wheel assemblies were positioned slightly farther outward than the front and middle wheels. The purpose of this change is to widen the rear portion of the rover's support polygon and increase the geometric lateral-stability margin.

The rear-wheel offset is a lateral positional change. It is not intended to create rear-wheel toe-out or camber. In the neutral flat-ground configuration:

- the rear wheel planes should remain parallel to the rover longitudinal centerline,
- the left and right rear-wheel offsets should be symmetric,
- the rear wheel axes should remain parallel to the corresponding front and middle wheel axes, and
- the track width should be measured between wheel centerplanes rather than outer grouser edges.

The original Week 4 track-width target was approximately:

$$W_{\text{track,baseline}} \approx 1.8 \text{ m.} \quad (22)$$

The exact rear offset should be measured from the completed CAD assembly. A conceptual sensitivity range of 50 mm to 100 mm per side can be used to estimate the additional torsional load introduced by the offset.

Using the Week 5 approximate lunar normal load per wheel [3]:

$$N_{\text{wheel}} = 267.3 \text{ N,} \quad (23)$$

the offset-induced moment is:

$$M_{\text{offset}} = N_{\text{wheel}} e_{\text{rear}}. \quad (24)$$

For an illustrative offset of 50 mm:

$$M_{\text{offset},50} = (267.3)(0.050) = 13.4 \text{ N m.} \quad (25)$$

For an illustrative offset of 100 mm:

$$M_{\text{offset},100} = (267.3)(0.100) = 26.7 \text{ N m.} \quad (26)$$

Applying the Week 5 design factor of 2.0 gives:

$$M_{\text{design},50} = 2(13.4) = 26.8 \text{ N m,} \quad (27)$$

$$M_{\text{design},100} = 2(26.7) = 53.4 \text{ N m.} \quad (28)$$

These calculations do not validate the rear bogie or wheel mounts. They indicate that the offset should remain moderate and that the rear suspension members should eventually be checked for combined bending and torsion.

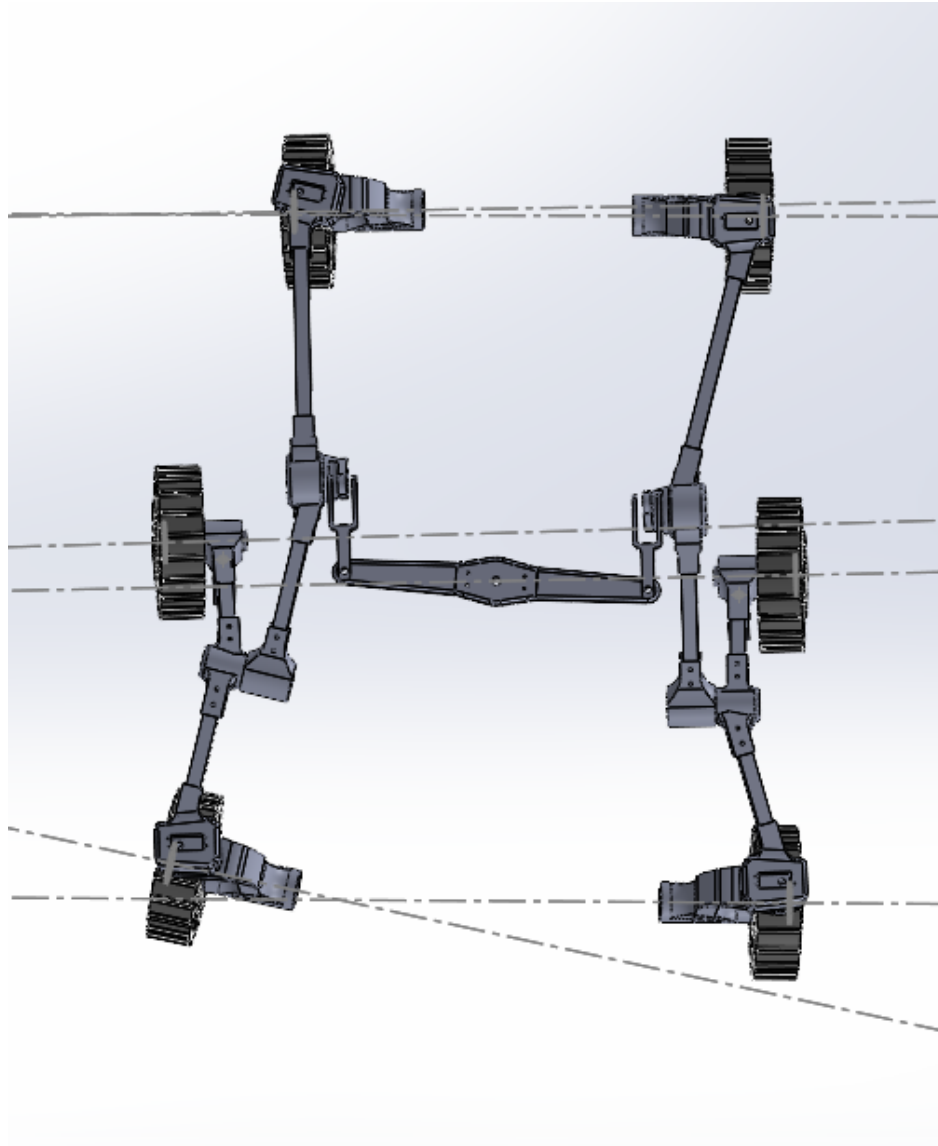


Figure 8: Top view showing the intentional outward lateral offset of the rear wheel assemblies.

14 Assembly Mates and Intended Motion

The beam-to-socket connections are treated as rigid connections. The rocker-to-bogie and main rocker pivot locations must remain rotational connections.

The assembly mating strategy includes:

- concentric mates between pivot pins and pivot bores,
- width or distance mates to control axial joint position,
- unlocked rotation at the rocker-to-bogie pivot,
- rigid or coincident mates at beam-to-socket connections,
- concentric wheel-hub and wheel-mount axes,
- parallel wheel planes in the neutral configuration,
- symmetric left and right suspension positioning, and
- equal wheel-axis height above a flat ground plane.

For a neutral flat-ground position, all equal-diameter wheel axes should be located the same vertical distance above the ground. If the wheel axes have different heights, the rocker and bogie angles should be adjusted rather than changing the wheel dimensions.

The assembly should also be checked through its available suspension motion to confirm that:

1. the wheel grousers do not contact the upper wheel mounts,
2. the rectangular members do not contact the pivot housings,
3. adjacent wheel mounts and beams do not interfere,
4. the center pivots retain axial clearance, and
5. the wheels can contact uneven terrain without overconstraining the assembly.

15 Completed Week 6 Mobility Assembly

The completed Week 6 model contains:

- six previously designed lunar wheels,
- left and right main rocker assemblies,
- left and right bogie assemblies,
- two detailed rocker-to-bogie pivot joints,

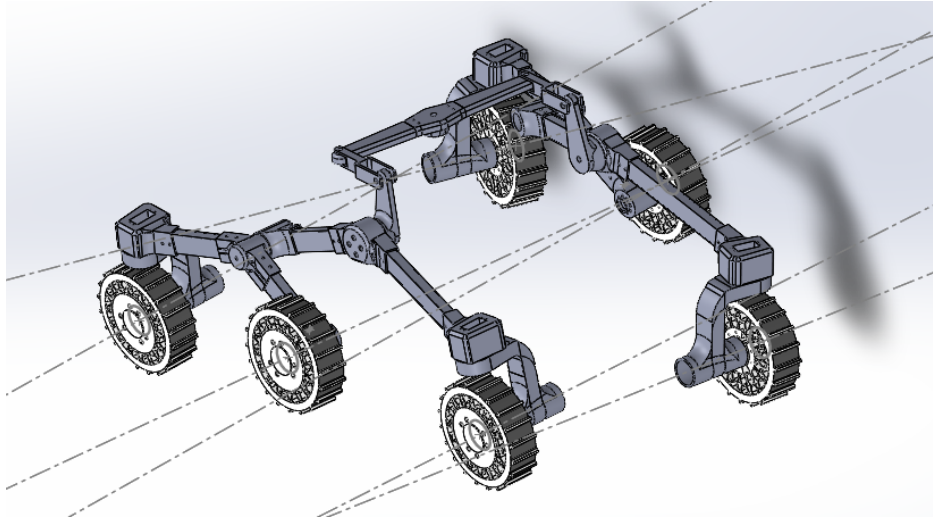


Figure 9: Completed Week 6 six-wheel rocker-bogie mobility assembly.

- 450 mm short bogie members,
- 500 mm long bogie members,
- 800 mm main rocker members,
- six detailed wheel-mount assemblies,
- rectangular socket connections,
- modeled through-bolt holes,
- an intentional rear-wheel lateral offset, and
- a preliminary differential or cross-link arrangement between the left and right sides.

The completed model represents the mobility subsystem only. The final chassis, dump bed, battery, electronics, sensors, motor internals, wiring, actuators, and complete rover packaging are intentionally excluded from the Week 6 scope.

16 Limitations

The Week 6 CAD model is a preliminary conceptual design and has the following limitations:

- The 100 mm by 50 mm suspension sections have not been verified through structural analysis.
- The preliminary 5 mm beam wall thickness has not been checked for bending stress, torsion, local buckling, or fatigue.

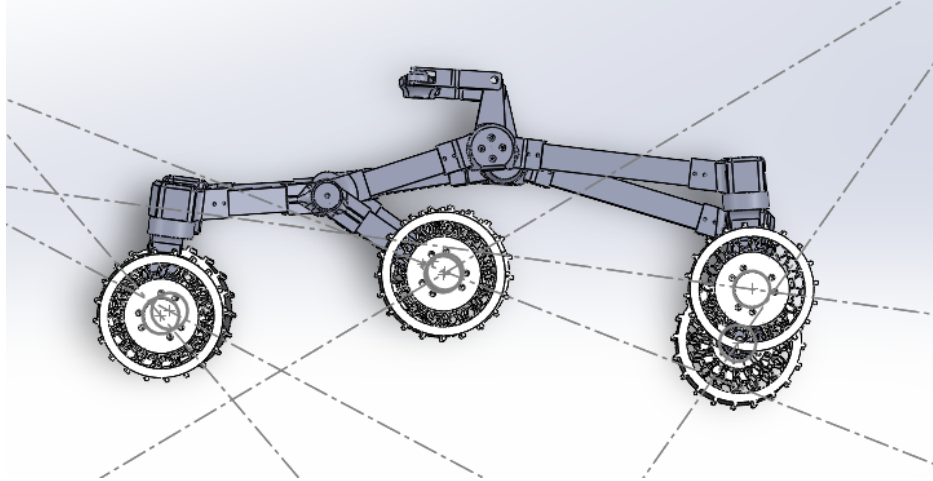


Figure 10: Side view of the completed Week 6 rocker-bogie suspension assembly.

- The 450 mm, 500 mm, and 800 mm member lengths were selected through CAD packaging and geometry refinement rather than structural optimization.
- The detailed final wheel-center spacing must be measured from the neutral-position assembly because the sockets, mounts, and pivots add effective length.
- The pivot pins, bushings, bearings, and fasteners have not been checked for shear stress, bearing stress, contact pressure, fatigue, or wear.
- Material selections have not been finalized.
- Welds, bonding, inserts, retaining rings, seals, and manufacturing tolerances are not fully modeled.
- The model does not include detailed motor, gearbox, bearing, encoder, or brake geometry.
- Lunar vacuum, dust, radiation, lubrication, thermal cycling, and thermal expansion are not included in the current model.
- The suspension has not been analyzed for one-wheel obstacle loading, wheel impact, asymmetric loading, or dynamic motion.
- The wider rear-wheel position increases torsional loading in the rear bogie members and wheel mounts.
- The exact rear-wheel offset remains subject to measurement and final chassis packaging.
- The current differential or cross-link system remains preliminary.
- The complete chassis, dump bed, battery, electronics, sensors, and full rover body are not included in the Week 6 model.

- The GrabCAD geometry was used as a reference and was not independently validated as a flight-qualified or manufacturing-ready design.

17 Engineering Summary

Week 6 converted the lunar regolith rover suspension from a simplified block model into a detailed six-wheel rocker-bogie CAD assembly. The imported GrabCAD assembly provided a useful mechanical reference, but the original geometry was substantially modified to match the current rover dimensions and suspension requirements.

The primary Week 6 design results were:

- replacement of the original circular rods with 100 mm by 50 mm rectangular hollow members,
- use of a preliminary 5 mm beam wall thickness,
- selection of a 450 mm short bogie member,
- selection of a 500 mm long bogie member,
- selection of an 800 mm main rocker member,
- development of 120 mm by 70 mm rectangular sockets,
- use of 104 mm by 54 mm socket openings,
- provision of approximately 2 mm beam clearance per side,
- use of approximately 8 mm socket walls,
- use of M12 through-bolt connections with approximately 13 mm holes,
- development of an approximately 30 mm central rocker-to-bogie pivot,
- redesign and lengthening of the wheel mounts for tire clearance,
- creation of a complete left and right six-wheel mobility assembly, and
- introduction of a modest outward rear-wheel offset to widen the rear portion of the support polygon.

The assembly is sufficiently detailed for the Week 6 conceptual CAD milestone. It demonstrates the suspension architecture, component proportions, connection methods, pivot locations, wheel mounts, selected member lengths, and geometry refinements.

The Week 6 model should not be described as structurally validated. It is a preliminary mobility assembly that provides a consistent foundation for the later full rover, chassis, and dump-bed assembly.

References

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